



ZAHIN UKASYAH

BACHELOR IN SOFTWARE ENGINEERING
@UNIVERSITI TEKNOLOGI MALAYSIA



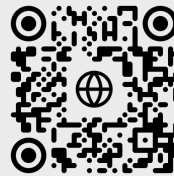
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Maluri, Kuala Lumpur



AI/ML Engineer specializing in NLP and Agentic RAG architectures. Proven ability in optimizing high-performance model and building scalable full-stack intelligent applications using React and Next.js. Dedicated to delivering high enterprise-grade AI solutions for the Maybank AI Center of Excellence (CoE)

SKILL

AI & MACHINE LEARNING

- **Orchestration:** Agentic RAG Pipelines, Multi-agent Systems, Semantic Search, Reranking.
- **Speech & Audio:** Speech-to-Speech (S2S), ASR, TTS, WebSockets.
- **Reliability & Ethics:** LLM Evaluation (RAGAS), AI Guardrails, Hallucination Mitigation, AI Observability.
- **ML Frameworks:** PyTorch, CNN, Scikit-learn, Fuzzy Logic.
- **Optimization:** FastAPI Async, Model Performance Tuning & Quantization.
- **Data & Vectors:** Qdrant, Supabase (pgvector), Hybrid Search, Pydantic.

FULL-STACK & SYSTEMS

- **Frontend:** React, Next.js, TypeScript, Tailwind CSS, Shadcn-UI.
- **Backend:** FastAPI, Node.js, RESTful & WebSocket, WebRTC API Design.
- **Mobile & Networking:** Expo (React Native), nGrok, HTTPS (Self-Cert).
- **Database & Cache:** PostgreSQL, MongoDB, Redis (Cache & Queuing).

DEVOPS & PROFESSIONAL

- **Infrastructure:** Docker, Linux (Ubuntu), Git/GitHub/GitHub Action.
- **Domain Knowledge:** FinTech AI Applications, KYC/Document AI.
- **Languages:** Bahasa Melayu (Native), English (Professional Efficiency).

EXPERIENCE

AI/ML ENGINEER CONTRACTOR (R&D) | MIMOS BERHAD

MARCH 2026 - JULY 2026

- Engineer a dual-pipeline architecture separating the conversational LLM from a background telemetry classifier, for real-time, hallucination-free emotion detection.
- Solve complex Manglish context-switching challenges by developing a dynamic intent router that analyze slang density on-the-fly to inject contextually relevant prompts.
- Lead the live Proof of Concept (PoC) deployment for pilot users, establishing a robust data privacy framework with anonymized longitudinal emotional telemetry tracking.

AI/ML ENGINEER TRAINEE (R&D) | MIMOS BERHAD

OCTOBER 2025 - FEBRUARY 2026

- Optimized real-time conversational flow and minimized core model memory by deploying a heavily quantized LLM.
- Ensured the factual integrity of therapeutic guidance by integrating a hybrid RAG pipeline, grounding AI responses securely in Malaysian clinical manuals and research papers.
- Enhanced multimodal user accessibility by integrating self-hosted Speech-to-Text (STT) and Text-to-Speech (TTS) models into a cross-platform React Native mobile app.

RESEARCH ASSISTANT | UNIVERSITI TEKNOLOGI MALAYSIA

FEBRUARY 2025 - JULY 2025

- Conducted systematic literature reviews (SLR) to identify methodologies in spatial-temporal deepfake detection.
- Curated and summarized high-impact research papers and datasets (e.g., FaceForensics++, Celeb-DF).
- Assisted in the acquisition and organization of diverse research materials, ensuring all frameworks and technical documentations were accessible for model development.
- Prepared detailed academic summaries and briefings.

REFERENCE

DR. HALINAWATI BINTI HIROL

Senior Lecturer,
System Electronic Engineering,
Faculty Malaysia-Japan International
Institute of Technology,
Universiti Teknologi Malaysia

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PUBLICATION

PRIMARY AUTHOR ON SOVEREIGN AI FOR MENTAL HEALTH: A PRIVACY-FIRST RAG ARCHITECTURE FOR LOW-RESOURCE LANGUAGES IN SENSITIVE DOMAINS.

- STATUS: **SUBMITTED MANUSCRIPT**
- AFFILIATION: **MIMOS BERHAD**
- CONFERENCE: **PRESENT @AIDI 2026, JUNE 24TH, 2026**



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PROJECTS

FULL-DUPLEX CONVERSATIONAL AI | MIMOS

- Developed a real-time Speech-to-Speech (S2S) system allowing simultaneous listening and speaking (Full-Duplex) based on PersonaPlex.
- Eliminated pipeline bottlenecks to achieve ultra-low sub-300ms latency.
- Optimized transformers to handle natural dialogue, including interruptions and barge-ins

MENTAL HEALTH MOBILE APP | MIMOS, MOSTI, JPA

- Architected a cross-platform mobile app using Expo for AI-driven emotional support.
- Integrated NLP-based Conversational AI to deliver personalized CBT-based interventions.
- Developed a Sentiment Analysis engine to track mood patterns and provide health suggestions.

VOICE CHATBOT FOR CALL CENTRE | RANHILL SAJ

- Developed a simple prototype for voice-based chatbot prototype for utility service inquiries.
- Integrated basic ASR and TTS components to facilitate voice-to-text-to-voice communication.
- Implemented intent recognition to handle queries such as bill checking and service status.

VOICE CLONING & SPEECH GENERATION | MIMOS

- Built a local-first Text-To-Speech engine powered by Qwen3 with a dual-track acoustic architecture.
- Enabled zero-shot voice cloning using only 3 sec(s) of reference audio with high speaker similarity.
- Integrated Instruction-Driven Design to generate custom personas via natural language prompts. (e.g., "A professional banker with a warm tone")

WATER METER OCR SCANNER | AIR SELANGOR

- Engineered a high-precision OCR pipeline using YOLOv8 for meter screen localization.
- Integrated Nanonets API for robust character extraction from low-quality/angled images.
- Automated manual utility readings into a digital dashboard for real-time tracking.

SPEECH-RAG PIPELINE (META SONAR) | MIMOS

- Developed an end-to-end Retrieval-Augmented Generation (RAG) pipeline that retrieves text-based knowledge directly from speech queries.
- Leveraged Meta SONAR to map speech and text into a unified embedding space for seamless cross-modal retrieval for Speech-to-Speech (S2S)

INVOLVEMENT

MALAYAN BANKING BERHAD

- **Academic Scholar (Current):** Awarded for academic excellence and leadership potential.
- **Maybank Ambassador (2024):** Represented Maybank to promote its initiatives by participated in Maybank events by organizer.

SOFTWARE ENGINEERING SOCIETY UTM

- **President (2025):** Led the society and oversee organization to benefit student community.
- **Director (2025):** Organized and directed society Annual Grand Meeting 2025 for the interim.
- **Secretary (2024):** Managed records and facilitated communication in the organization.

PEMBIMBING RAKAN SISWA UTM

- **Acting President (2024):** Led the organization objectives and ensured smooth operations.
- **Vice President (2024):** Assisted in leadership, planning, and organizational management.

SOFTWARE ENGINEERING DEPARTMENT UTMKL

- **Award Recipient (2025):** Recipient for the Undergraduate Research Opportunity (UROP) on Biometric Image Iris Recognition research (FYP)

STUDENT REPRESENTATIVE COUNCIL UTM

- **Exco Student Welfare (2024):** Advocated for student welfare and organized support initiatives.
- **Director (2024):** Initiate the Scholarship Hunter project to aid student on financial issues.
- **Deputy Director (2024):** Organized the planning for Event: Eid Al-Fitr Celebration UTMKL 2024.
- **Deputy Director (2024):** Organized the planning for Event: Minggu Mesra Mahasiswa 2024.
- **Liaison Officer (2024):** Facilitated communication with China's delegates on UTMKL business travel.

INSTITUTIONAL ENGINEERING MALAYSIA (IEM)

- **Student Advisor (2026):** Advocated for guiding and assisting committee to achieve objectives.